Infinite series

Lecture

Series

let's start with a sequence $\{\boldsymbol{a_n}\}_{n=1}^{\infty}$. and define the following,

$$s_1 = \boldsymbol{a_1}$$

$$s_2 = \boldsymbol{a_1} + \boldsymbol{a_2}$$

$$s_3 = \boldsymbol{a_1} + \boldsymbol{a_2} + \boldsymbol{a_3}$$

$$s_n = \boldsymbol{a_1} + \boldsymbol{a_2} + \boldsymbol{a_3} + \cdots + \boldsymbol{a_n} = \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \boldsymbol{a_i}$$

The s_n are called **partial sums** and notice that they will form a sequence, $\{\boldsymbol{s_n}\}_{n=1}^{\infty}$.

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \sum_{i=1}^{n} a_i = \sum_{i=1}^{\infty} a_i$$

If the sequence of partial sums, $\{s_n\}_{n=1}^{\infty}$, is convergent and its limit is finite then we also call the infinite series, $\sum_{i=1}^{\infty} a_i$ **convergent** and if the sequence of partial sums is divergent then the infinite series is also called **divergent**.

Now, in $\sum_{i=1}^{\infty} a_i$ the i is called the **index of summation** or just **index** for short and note that the letter we use to represent the index does not matter. So for example the following series are all the same.

$$\sum_{i=1}^{\infty} \frac{3}{i^2+1} = \sum_{k=1}^{\infty} \frac{3}{k^2+1} = \sum_{n=1}^{\infty} \frac{3}{n^2+1} \qquad etc.$$

Properties

If $\sum \boldsymbol{a_n}$ and $\sum \boldsymbol{b_n}$ are both convergent series then,

1. $\sum \boldsymbol{c\, a_n}$, where c is any number, is also convergent and $\sum \boldsymbol{c\, a_n} = \boldsymbol{c} \sum \boldsymbol{a_n}$

2. $\sum_{n=k}^{\infty} \boldsymbol{a_n} \pm \sum_{n=k}^{\infty} \boldsymbol{b_n}$ is also convergent and,

$$\sum_{n=k}^{\infty} \boldsymbol{a_n} \pm \sum_{n=k}^{\infty} \boldsymbol{b_n} = \sum_{n=k}^{\infty} (\boldsymbol{a_n} \pm \boldsymbol{b_n})$$

$$\sum_{n=0}^{\infty} \boldsymbol{a_n} \sum_{n=0}^{\infty} \boldsymbol{b_n} \neq \sum_{n=0}^{\infty} (\boldsymbol{a_n} . \boldsymbol{b_n})$$

index shift

The basic idea behind index shifts is to start a series at a different value

Consider the following series,

$$\sum_{n=2}^{\infty} \frac{n+5}{2^n}$$

Suppose that for some reason we wanted to start this series at $n = 0$, but we didn't want to change the value of the series. This means that we can't just change the $n = 2$ to $n = 0$ as this would add in two new terms to the series and thus change its value.

Performing an index shift is a fairly simple process to do. We'll start by defining a new index, say i, as follows, $i=n=2$.

We can now completely rewrite the series in terms of the index i instead of the index n simply by plugging in our equation for n in terms of i.

$$\sum_{n=2}^{\infty} \frac{n+5}{2^n} = \sum_{i=0}^{\infty} \frac{(i+2)+5}{2^{i+2}} = \sum_{i=0}^{\infty} \frac{i+7}{2^{i+2}}$$

we'll change the final i back into an n to get,

$$\sum_{n=2}^{\infty} \frac{n+5}{2^n} = \sum_{n=0}^{\infty} \frac{n+7}{2^{n+2}}$$

$$\sum_{n=2}^{\infty} \frac{n+5}{2^n} = \frac{7}{2^2} + \frac{8}{2^3} + \frac{9}{2^4} + \frac{10}{2^5} + \cdots$$

$$\sum_{n=0}^{\infty} \frac{n+7}{2^{n+2}} = \frac{7}{2^2} + \frac{8}{2^3} + \frac{9}{2^4} + \frac{10}{2^5} + \cdots$$

Convergence/Divergence of Series

If the sequence of partial sums is a convergent sequence (*i.e.* its limit exists and is finite) then the series is also called **convergent** and in this case if $\lim_{n\to\infty} \boldsymbol{s_n} = s$ then, $\sum_{\boldsymbol{i=1}}^{\infty} \boldsymbol{a_i} = \boldsymbol{s}$. Likewise, if the sequence of partial sums is a divergent sequence (*i.e.* its limit doesn't exist or is plus or minus infinity) then the series is also called **divergent.**

Example 1.

Determine if the following series is convergent or divergent. If it converges determine its value. $\sum_{n=1}^{\infty} \boldsymbol{n}$.

Solution:

$$s_n = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\lim_{n\to\infty} \frac{n(n+1)}{2} = \infty$$

Therefore, the sequence of partial sums diverges to ∞ and so the series also diverges.

Example 2.

Determine if the following series converges or diverges. If it converges determine its sum.

$\sum_{n=0}^{\infty}(-1)^n$.

$$s_0 = 1$$

$$s_1 = 1 - 1 = 0$$

$$s_2 = 1 - 1 + 1 = 1$$

$$s_3 = 1 - 1 + 1 - 1 = 0$$

$$.$$

$$etc.$$

So, it looks like the sequence of partial sums is,

$$\{s_n\}_{n=0}^{\infty} = \{1, 0, 1, 0, 1, 0, 1, 0, 1, \dots\}$$

and this sequence diverges since $\lim_{n\to\infty} s_n$ doesn't exist. Therefore, the series also diverges.

Divergence Test

If $\lim_{n\to\infty} a_n \neq 0$ then $\sum a_n$ will diverge.

Example

Determine if the following series is convergent or divergent

$$\sum_{n=0}^{\infty} \frac{4n^2 - n^3}{10 + 2n^3}$$

Solution:

That's what we'll do here.

$$\lim_{n\to\infty} \frac{4n^2 - n^3}{10 + 2n^3} = -\frac{1}{2} \neq 0$$

The limit of the series terms isn't zero and so by the Divergence Test the series diverges.

Special Series:

1) **Geometric Series**

A geometric series is any series that can be written in the form,

$$\sum_{n=1}^{\infty} ar^{n-1}$$

or, with an index shift the geometric series will often be written as,

$$\sum_{n=0}^{\infty} ar^{n}$$

If we start with the first form it can be shown that the partial sums are,

$$s_n = \frac{a(1 - r^n)}{1 - r}$$

Therefore, a geometric series will converge if $-1<r<1$, which is usually written $|r|<1$, its value is,

$$\sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^n = \frac{a}{1 - r}$$

$$a)\sum_{n=1}^{\infty} 9^{-n+2}\ 4^{n+1}$$

$$\sum_{n=1}^{\infty} 9^{-n+2}\ 4^{n+1} = \sum_{n=1}^{\infty} \frac{4^{n+1}}{9^{n-2}} = \sum_{n=1}^{\infty} \frac{4^2 4^{n-1}}{9^{-1} 9^{n-1}} = \sum_{n=1}^{\infty} 144\left(\frac{4}{9}\right)^{n-1}$$

So, this is a geometric series with a =144 and $r = \frac{4}{9}$ <1.

Therefore, since $|r|$ <1

we know the series will converge and its value will be,

$$\sum_{n=1}^{\infty} 9^{-n+2}\ 4^{n+1} = \frac{144}{1-\frac{4}{9}}$$

$$b)\sum_{n=0}^{\infty}\frac{(-4)^{3n}}{5^{n-1}}$$

$$\sum_{n=0}^{\infty}\frac{(-4)^{3n}}{5^{n-1}}=\sum_{n=0}^{\infty}\frac{((-4)^{3})^{n}}{5^{n}\,5^{-1}}$$

$$=\sum_{n=0}^{\infty}5\frac{(-64)^{n}}{5^{n}}=\sum_{n=0}^{\infty}5\left(\frac{-64}{5}\right)^{n}$$

So, we've got it into the correct form and we can see that $a = 5$ and $r = \frac{-64}{5}$

Also note that $|r| \geq 1$ and so this series diverges.

Harmonic Series

Harmonic series can be written in the form $\sum_{n=1}^{\infty} \frac{1}{n}$

The harmonic series is divergent

Example 1: Show that each of the following series are divergent.

1) $\sum_{n=1}^{\infty} \frac{5}{n}$

2) $\sum_{n=4}^{\infty} \frac{1}{n}$

$$\sum_{n=1}^{\infty} \frac{5}{n} = 5 \sum_{n=1}^{\infty} \frac{1}{n}$$

Now, is divergent and so five times this

it will still be divergent.

$$2)\ \sum_{n=4}^{\infty} \frac{1}{n}$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \sum_{n=4}^{\infty} \frac{1}{n}$$

$$\Rightarrow \quad \sum_{n=4}^{\infty} \frac{1}{n} = \sum_{n=1}^{\infty} \frac{1}{n} - \frac{11}{6}$$

In this case we are subtracting a finite number from a divergent series.

This subtraction will not change the divergence of the series

Therefore, this series is divergent.

So, some general rules about the **convergence/divergence** of a series are now in order. Multiplying a series by a constant will not change the convergence/divergence of the series and adding or subtracting a constant from a series will not change the convergence/divergence of the series.

Fact (The p –series Test)

If $k > 0$ then $\sum_{n=k}^{\infty}\left(\frac{1}{n^p}\right)$ converges if $p>1$ and diverges if $p\leq 1$.

Example 1: Determine if the following series is convergent or divergent.

$$1)\ \sum_{n=4}^{\infty}\left(\frac{1}{n^7}\right) \qquad\qquad 2)\ \sum_{n=1}^{\infty}\left(\frac{1}{\sqrt{n}}\right)$$

Solution:

1) In this case $p = 7 > 1$, and so by this fact the series is convergent.

2) For this series, $p = \frac{1}{2} < 1$ and so the series is divergent by the fact.

⊗ *Alternating Series Test*

An **alternating series** is any series, $\sum a_n$, for which the series terms can be written in one of the following two forms.

$$a_n = (-1)^n b_n \qquad b_n \geq 0$$

$$a_n = (-1)^{n+1} b_n \qquad b_n \geq 0$$

There are many other ways to deal with the alternating sign, but they can all be written as one of the two forms above. For instance,

$$(-1)^{n+2} = (-1)^n(-1)^2 = (-1)^n$$

$$(-1)^{n-1} = (-1)^{n+1}(-1)^{-2} = (-1)^{n+1}$$

Alternating Series Test

Suppose that we have a series $\sum a_n$ and either $a_n = (-1)^n b_n$ or $a_n = (-1)^{n+1} b_n$ where $b_n \geq 0$ for all n. Then if,

1. $\mathbf{\lim_{n\to\infty} b_n = 0}$ **and,**
2. $\{b_n\}$ is a decreasing sequence

the series $\sum a_n$ is convergent.

Example 1: Determine if the following series is convergent or divergent.

$$\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}}{n}\right)$$

Solution:

First, identify the b_n for the test.

$$\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}}{n}\right)=\sum_{n=1}^{\infty}(-1)^{n+1}\left(\frac{1}{n}\right), \qquad b_n=\frac{1}{n}$$

$$\lim_{n\to\infty} b_n=\lim_{n\to\infty}\frac{1}{n}=0$$

$$b_n=\frac{1}{n}>\frac{1}{n+1}=b_{n+1}$$

Both conditions are met and so by the Alternating Series Test the series must converge.

Example 2: Determine if the following series is convergent or divergent.

$$\sum_{n=1}^{\infty}\left(\frac{(-1)^n n^2}{n^2+5}\right)$$

Solution

First, identify the b_n for the test.

$$\sum_{n=1}^{\infty}\left(\frac{(-1)^n n^2}{n^2+5}\right)=\sum_{n=1}^{\infty}(-1)^{n+1}\left(\frac{n^2}{n^2+5}\right)$$

$$b_n=\frac{n^2}{n^2+5}.$$

$$\lim_{n\to\infty} b_n=\lim_{n\to\infty}\frac{n^2}{n^2+5}=1\neq 0$$

So, the first condition isn't met and so there is no reason to check the second.

Example 3:

Determine if the following series is convergent or divergent.

$$\sum_{n=2}^{\infty}\left(\frac{\cos n\pi}{\sqrt{n}}\right)$$

Solution:

$\cos(n\pi) = (-1)^n$

$$\sum_{n=1}^{\infty}\left(\frac{\cos n\pi}{\sqrt{n}}\right) = \sum_{n=1}^{\infty}(-1)^n\left(\frac{1}{\sqrt{n}}\right), \qquad b_n = \frac{1}{\sqrt{n}}$$

Checking the two condition gives,

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty}\frac{1}{\sqrt{n}} = 0$$

$$b_n = \frac{1}{\sqrt{n}} > \frac{1}{\sqrt{n+1}} = b_{n+1}$$

The two conditions of the test are met and so by the Alternating Series Test the series is convergent.

Ratio Test

Suppose we have the series $\sum a_n$. Define,

$$\boldsymbol{L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|}$$

Then,

1. if $L<1$ the series is absolutely convergent (and hence convergent).
2. if $L>1$ the series is divergent.
3. if $L=1$ the series may be divergent, conditionally convergent, or absolutely convergent.

Example 1: Determine if the following series is convergent or divergent.

$$\sum_{n=0}^{\infty}\left(\frac{n!}{5^n}\right)$$

Solution:

$a_n = \frac{n!}{5^n}$ $\qquad$ $a_{n+1} = \frac{(n+1)!}{5^{n+1}}$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{5^{n+1}} \times \frac{5^n}{n!} = \frac{(n+1)n!}{5^n\,5} \times \frac{5^n}{n!} = \frac{n+1}{5}$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{n+1}{5}\right| = \frac{n+1}{5}$$

$$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n+1}{5} = \infty > 1$$

So, by the Ratio Test this series diverges.

Example 2: Determine if the following series is convergent or divergent.

$$\sum_{n=2}^{\infty}\left(\frac{n^2}{(2n-1)!}\right)$$

Solution:

$$a_n = \frac{n^2}{(2n-1)!} \qquad a_{n+1} = \frac{(n+1)^2}{(2n+1)!}$$

$$\frac{a_{n+1}}{a_n} = \frac{n^2+2n+1}{(2n+1)!} \times \frac{(2n-1)!}{n^2}$$

$$= \frac{n^2+2n+1}{(2n+1)(2n)(2n-1)!} \times \frac{(2n-1)!}{n^2} = \frac{n^2+2n+1}{4n^4+2n^3}$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{n^2+2n+1}{4n^4+2n^3}\right| = \frac{n^2+2n+1}{4n^4+2n^3}$$

$$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n^2+2n+1}{4n^4+2n^3} = 0 < 1$$

So, by the Ratio Test this series converges absolutely and so converges.

Example 4: Determine if the following series is convergent or divergent.

$$\sum_{n=1}^{\infty}\left(\frac{(n!)^2}{(2n)!}\right)$$

Solution:

$a_n = \frac{(n!)^2}{(2n)!}$ $\qquad$ $a_{n+1} = \frac{((n+1)!)^2}{(2n+2)!}$

$$\frac{a_{n+1}}{a_n} = \frac{((n+1)!)^2}{(2n+2)!} \times \frac{(2n)!}{(n!)^2} = \frac{(n+1)^2(n!)^2}{(2n+2)(2n+1)(2n)!} \times \frac{(2n)!}{(n!)^2}$$

$$= \frac{n^2+2n+1}{4n^2+6n+2}$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{n^2+2n+1}{4n^2+6n+2}\right| = \frac{n^2+2n+1}{4n^2+6n+2}$$

$$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n^2+2n+1}{4n^2+6n+2} = \frac{1}{4} < 1$$

So, by the Ratio Test this series converges absolutely and so converges.

Root Test

Suppose that we have the series $\sum a_n$. Define,

$$L = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} |a_n|^{\frac{1}{n}}$$

Then,

1. if $L<1$ the series is absolutely convergent (and hence convergent).
2. if $L>1$ the series is divergent.
3. if $L=1$ the series may be divergent, conditionally convergent, or absolutely convergent.

Fact

$$\lim_{n\to\infty} n^{\frac{1}{n}} = 1$$

Example 1: Determine if the following series is convergent or divergent.

$$\sum_{n=1}^{\infty}\left(\frac{n^n}{3^{2n+1}}\right)$$

Solution:

$$a_n = \frac{n^n}{3^{2n+1}} \qquad \sqrt[n]{|a_n|} = \sqrt[n]{\frac{n^n}{3^{2n+1}}} = \left(\frac{n^n}{3^{2n+1}}\right)^{\frac{1}{n}}$$

There really isn't much to these problems other than computing the limit and then using the root test. Here is the limit for this problem.

$$L = \lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\frac{n}{3^{2+\frac{1}{n}}} = \frac{\infty}{3^2} = \infty > 1$$

So, by the Root Test this series is divergent.

Example 2: Determine if the following series is convergent or divergent.

$$\sum_{n=0}^{\infty}\left(\frac{5n-3n^3}{7n^3+2}\right)^n$$

Solution:

$a_n = \left(\frac{5n-3n^3}{7n^3+2}\right)^n \qquad \sqrt[n]{|a_n|} = \left|\frac{5n-3n^3}{7n^3+2}\right|$

There really isn't much to these problems other than computing the limit and then using the root test. Here is the limit for this problem.

$$L = \lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\left|\frac{5n-3n^3}{7n^3+2}\right| = \left|\frac{-3}{7}\right| = \frac{3}{7} < 1$$

Therefore, by the Root Test this series converges absolutely and hence converges

thank you